

Agnieszka Żaba

I am a doctoral student in the AGH Environmental Physics Group. My research focuses on cloud microphysics modelling and climate-science applications of stable water isotopes. Research software engineering is a key part of my work. As an active member of the github.com/open-atmos community, I maintain the PySDM and the open-atmos-jupyter-utils Python packages.

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AgnieszkaZaba

Tech stack

Python

NumPy, SciPy, Matplotlib, Pandas, Pytest, Tensorflow, PyTorch, CUDA Python JIT

Git, Github Actions, SQL, C++

Mathematica, R, Matlab

Jupyter, L^AT_EX, BibT_EX

Languages

 Polish ● ● ● ● ●

 English ● ● ● ● ●

 Spanish ● ● ● ● ●

Certificates

NVIDIA DLI:

 Fundamentals of Deep Learning, Accelerated Computing with CUDA Python

Nature MasterClass:

 Writing a Research Paper, Publishing a Research Paper, Writing and Publishing a Review Paper

Hobby

Hiking, board games, escape rooms, logic puzzles, drawing

Education

AGH University of Krakow

Kraków, PL

10.2024 –
(ongoing)

Ph.D. in Physics
Faculty of Physics and Applied Computer Science
Modelling isotopic signatures in precipitation using particle-based cloud microphysics
Research advisors: Dr. Sylwester Arabas, Prof. Mirosław Zimnoch

07.2023 –
10.2021

MSc in Computational Mathematics
Faculty of Applied Mathematics
Finite difference scheme for multidimensional interdiffusion model
Research advisor: Prof. Bogusław Bożek

Wroclaw University of Science and Technology

Wrocław, PL

02.2018 –
10.2013

BSc in Engineering Applied Mathematics
Faculty of Mathematics
Statistical analysis of Lévy walks
Research advisor: Prof. Marcin Magdziarz

Conference Presentations

08.2026
(accepted)

AMS Madison Summit: 17th Conference of Cloud Physics Madison, WI, USA
Modeling Isotopic Fractionation with Particle-Based Microphysics Using the Bolin Number

08.2025

EuroSciPy <https://euroscipy.org/> AGH, Kraków, PL
Poster: *Can Your Research Notebook Survive a Code Update? Maintaining Reproducibility with CI and Automated Testing*

04.2025

Improving Scientific Software Boulder, CO, USA
UCAR Software Engineering Assembly

02.2025

Free & Open Source Developers' Euro Meeting Brussels, BE
Université Libre de Bruxelles
5 min talk: open-atmos-jupyter-utils

Teaching

06.2026 –
01.2026

Modelling of Physical Systems AGH, Kraków, PL
computational modelling lab classes with projects for first-year graduate students

02.2026 –
10.2025

Modelling of Environmental Transport Processes AGH, Kraków, PL
computational project classes for final-year undergrad students

02.2026 –
10.2025

Modeling of Flow Systems AGH, Kraków, PL
computational modelling lab for final-year undergrad students

03.2025 –
06.2025

Nano, Micro and Macro-Scales in Earth Systems AGH, Kraków, PL
computational modelling lab for first-year undergrad students

01.2025 –
10.2024

Environmental Transport Processes AGH, Kraków, PL
computational modelling lab for final-year undergrad students

Workshops & Summer Schools

26.06.2026 – 22.06.2026 (planned)	IsoSchool 2026 and International Clumped Isotope Workshop (ICIW10) Department of Earth Sciences at Utrecht University	Utrecht, Netherlands
26.09.2025 – 22.09.2025	Compact Course Atmospheric Chemistry and Dynamics <i>Energy and Climate Research at the Forschungszentrum Jülich Institute of Climate and Energy Systems (ICE)</i>	Jülich, FR
20.06.2025 – 15.06.2025	CoPartCoFlow <i>Complex particle transport in complex flows with environmental applications</i> CNRS - IRPHE & University of Wisconsin - Madison	La londe les Maures, FR

Community Service

02.2026	Volunteer at FOSDEM Conference	Brussels, BE
07.2025	Student Helper at Goldchmidt Conference	Prague, CZK
09.2015 – 09.2012	Scouts Team Leader	Opole, PL